SOFR Transition and USD Benchmark Rate Note

Reference note prepared as of 2026-06-30 for the credit and relative-value work on this engagement. It summarizes the USD LIBOR-to-SOFR transition, the mechanics of SOFR-based pricing now standard in floating-rate credit facilities, and what that means for reading this file's floating-rate instruments correctly. It is background/reference material and is not itself a source of any scored valuation input in this engagement's fixed-rate senior notes analysis.

1. Background: USD LIBOR Cessation

The UK Financial Conduct Authority ceased compelling panel-bank submission for USD LIBOR settings after end-June 2023, and ICE Benchmark Administration correspondingly ceased publication of the remaining USD LIBOR tenors (overnight/1-month/3-month/6-month/12-month) as representative rates after June 30, 2023. Some tenors continued briefly on a non-representative, 'synthetic' basis strictly for legacy contract runoff, but no new USD LIBOR exposure has been permissible since that date, and the synthetic publication itself has since lapsed. A document, model or trade ticket dated after June 30, 2023 that still prices new exposure off USD LIBOR is not describing a live market rate - it is either referencing history, referencing a legacy fallback that has already converted to SOFR-plus-spread-adjustment, or it is simply wrong. This is a real, externally verifiable transition, not an assumption specific to this engagement.

The Alternative Reference Rates Committee (ARRC), convened by the Federal Reserve Board and the Federal Reserve Bank of New York together with market participants, identified the Secured Overnight Financing Rate (SOFR) as its recommended alternative for USD LIBOR in 2017 and spent the subsequent several years publishing recommended fallback language, conventions and a phased adoption timeline for cash products - syndicated loans, floating-rate notes, securitizations and business loans - so that the transition away from LIBOR would not depend on last-minute, contract-by-contract renegotiation. The Federal Reserve's own SOFR page and the LIBOR Act (adopted by Congress in 2022 as part of the Consolidated Appropriations Act, 2023, and implemented by Federal Reserve regulation) supply a statutory SOFR-based replacement rate for legacy contracts that never included adequate fallback language of their own, closing the gap for the minority of instruments that would otherwise have had no clear successor rate at cessation.

2. SOFR: Definition and Publication

SOFR is a broad measure of the cost of borrowing cash overnight, collateralized by Treasury securities. It is published each business day by the Federal Reserve Bank of New York, based on transactions in the tri-party repo market, General Collateral Finance (GCF) repo, and a subset of bilateral Treasury repo transactions cleared through the Fixed Income Clearing Corporation (FICC). Because it is a secured, overnight, transaction-based rate, SOFR sits structurally below where an unsecured interbank term rate such as LIBOR would trade, all else equal - LIBOR embedded an unsecured bank-credit premium and a term-liquidity premium that a secured overnight repo rate simply does not carry. That structural gap is precisely what the ISDA/ARRC fallback spread adjustments (Section 4 below) exist to bridge for legacy contracts converting from LIBOR to SOFR, rather than leaving economically comparable counterparties worse or better off purely as an artifact of the benchmark switch.

SOFR is inherently an overnight rate with no built-in term structure, which created the central design problem for cash products that need a rate fixed in advance for a coupon period (a quarterly-pay floating-rate note, for instance): an overnight rate observed only in arrears does not, by itself, give a borrower or lender a known cash-flow amount until the accrual period is already over. Two distinct conventions solve this, described next.

3. Term SOFR vs. Compounded SOFR in Arrears

Term SOFR is a forward-looking rate for a stated tenor (commonly 1-month, 3-month, 6-month or 12-month), derived from SOFR futures and OIS market pricing and administered by CME Group Benchmark Administration Limited under a license the ARRC endorsed for use in cash products such as business loans and securitizations, subject to the ARRC's scope-of-use guidance discouraging its use in the deepest, most liquid derivatives markets in order to protect the futures and OIS market liquidity Term SOFR itself depends on. Term SOFR behaves, operationally, the way LIBOR did: it is set at the start of an interest period and the borrower knows the coupon amount immediately.

Compounded SOFR in arrears instead compounds the daily published overnight SOFR fixings over the accrual period itself, so the exact rate for a given interest period is only fully known a short lookback window before that period ends (commercial loan and note documentation commonly uses a 2- or 5-business-day lookback, sometimes with a lockout on the final days of the period so payment processing has a known number). Compounded-in-arrears SOFR is the ARRC's preferred, most robust convention because it is built directly from the published overnight rate with no separate derivatives-market administration layer, but it is operationally less convenient for a borrower that wants to know its coupon in advance, which is why Term SOFR remains the dominant convention in the syndicated and middle-market loan space this engagement's credit facility sits in.

4. ISDA/ARRC Fallback Spread Adjustments (Legacy Fallback Only)

For legacy USD LIBOR contracts that fell back to SOFR at cessation under ISDA's IBOR Fallbacks Supplement (and the closely parallel ARRC-recommended hardwired fallback language used in cash products), the fallback rate is Term SOFR or compounded SOFR in arrears plus a fixed spread adjustment, one per tenor, designed to approximate the historical median difference between that USD LIBOR tenor and the corresponding SOFR rate compounded in arrears over a five-year lookback. Those spread adjustments were fixed and published by Bloomberg Index Services Limited (as ISDA's designated vendor) once USD LIBOR's cessation date was confirmed, and - because they are fixed constants rather than a live series - they do not move again, unlike SOFR itself. Approximate published levels,

smallest to largest tenor: roughly 11.4 basis points for one-month, 26.2 basis points for three-month, 42.8 basis points for six-month, and 71.5 basis points for twelve-month exposure - the ordering (longer tenor, larger adjustment) reflects that a longer unsecured interbank term rate historically carried a larger term-and-credit premium over overnight secured funding.

The spread-adjustment mechanism matters only for a contract that was live under LIBOR at cessation and fell back mechanically; it is not part of the mechanics of a facility, like the one in this engagement's credit agreement, that was originated after the LIBOR-to-SOFR transition and priced on Term SOFR (or compounded SOFR) from day one, with no separate legacy-spread-adjustment leg to carry forward. A 2026-dated instrument citing a LIBOR-to-SOFR spread adjustment as a live, ongoing pricing input - rather than as historical background - would be describing something that does not apply to a facility that never priced off LIBOR in the first place.

5. Credit Agreement Mechanics: SOFR-Based Pricing Grids

A modern floating-rate credit facility (term loan or revolver) originated after mid-2023 is priced as Term SOFR for the selected interest-period tenor, subject to a stated SOFR floor (commonly zero, occasionally a small positive floor such as 0.50%, negotiated as part of the original pricing), plus an Applicable Margin that steps down (or up) across a leverage-based pricing grid, and typically also a small, fixed 'credit spread adjustment' or 'CSA' the arranger group negotiated into the facility's own pricing at origination - a design choice about that deal's economics, not a regulatory fallback mechanic, and conceptually distinct from the ISDA/ARRC legacy fallback spread adjustments described in Section 4, even though both are commonly abbreviated 'CSA' and are easy to conflate. Some post-transition facilities price 'SOFR flat' (Term SOFR plus margin, no separate CSA at all); others retain a small CSA as a negotiated feature. Reading a credit agreement's own defined terms for 'Term SOFR', 'Adjusted Term SOFR' and 'Applicable Margin' is the only reliable way to know which convention a specific facility actually uses - the market convention has not fully standardized on one approach.

For this engagement, the practical consequence is narrow but important: any floating-rate exposure in the target's capital structure needs to be priced off the genuine, current SOFR curve (Term SOFR for the relevant tenor, or the relevant SOFR average), never off a LIBOR curve, and never off the legacy fallback spread adjustments in Section 4, which do not apply to a facility that was never a LIBOR facility. The target's own senior notes are fixed-rate and are unaffected by any of this; the benchmark-rate mechanics in this note bear only on the secured credit facility side of the capital structure and on any relative-value comparison that needs a floating-rate discount input.

6. Relevance to Relative-Value Analysis

Benchmark-rate mechanics do not change how a fixed-rate corporate bond is priced or discounted - a fixed coupon, fixed-maturity instrument's yield-to-worst and spread analysis runs entirely off the nominal Treasury curve and the swap curve, neither of which is affected by whether a given borrower's separate floating-rate facility happens to be priced on SOFR, a legacy LIBOR fallback, or something else. Where benchmark-rate mechanics do matter is in any leverage or coverage calculation that nets in the cost of the floating-rate piece of the capital structure, and in reading covenant definitions (interest coverage ratios, fixed-charge coverage ratios) that reference a facility's own defined 'Adjusted Term SOFR' or equivalent term - a reader who substitutes a stale LIBOR-era assumption for that defined term would misstate projected debt service on the floating-rate piece of the structure, even though the fixed-rate notes analysis itself is entirely unaffected.

7. Key Dates

For quick reference: ARRC recommended SOFR as its preferred alternative in 2017; the Federal Reserve Board and the New York Fed began publishing SOFR in April 2018; CME Term SOFR launched and ARRC formally recommended its use in certain cash products in 2021; the ISDA IBOR Fallbacks Supplement and Protocol took effect in January 2021, fixing the spread adjustments described in Section 4 shortly after; most USD LIBOR settings ceased after December 31, 2021, but the handful of USD tenors used most heavily in cash products (overnight and 1-, 3-, 6- and 12-month) continued, briefly on a non-representative synthetic basis, until final cessation after June 30, 2023 - the date this note and every 2026-dated document in this engagement treats as the bright line after which USD LIBOR is not a live benchmark under any circumstance.

8. The 'SOFR First' Initiative and Market Adoption Sequencing

Adoption did not happen uniformly across products on a single date. The CFTC's Market Risk Advisory Committee endorsed a phased 'SOFR First' initiative recommending that interdealer brokers first move their default trading conventions to SOFR in the linear interest-rate swap market (July 2021), before sequentially extending the same convention change to cross-currency swaps, non-linear derivatives, and exchange-traded futures and options over the following months. That sequencing mattered because swap-market liquidity in SOFR is itself one of the inputs CME's Term SOFR calculation depends on; moving the most liquid interdealer swap flow to SOFR first was a deliberate way to deepen the SOFR swap and futures markets ahead of the cash-product cutover, so that Term SOFR itself would be soundly built by the time loan and bond markets needed to rely on it as a rate-setting mechanism. The cash-lending market's own adoption followed a similar staged pattern: new-issue leveraged loans priced on Term SOFR well before the final USD LIBOR cessation date, so that by the time this engagement's own credit facility was negotiated, Term SOFR-based pricing was already the uncontested market standard rather than a novel convention.

9. LIBOR-SOFR Basis Behavior During Market Stress

One reason the transition mattered economically, not just administratively, is that the LIBOR-SOFR spread is not

constant - it widens during periods of bank funding stress and narrows in calm markets, because LIBOR embeds unsecured interbank credit risk that a secured, Treasury-collateralized repo rate like SOFR does not carry. The most widely cited illustration is March 2020: as the onset of pandemic-related market stress drove a flight to the safety and liquidity of Treasury collateral, SOFR briefly fell as repo rates richened, while USD LIBOR rose as unsecured bank funding costs widened - the two benchmarks moved in opposite directions at the exact moment a floating-rate borrower's payment obligation would have mattered most. A fixed spread adjustment set once, as the ISDA/ARRC fallback spread adjustments were, cannot track that kind of time-varying basis; it is a reasonable point-in-time approximation of the historical median relationship, not a hedge against a future stress episode reproducing the same dynamic. This is one of the reasons the ARRC consistently described the fallback spread adjustment as a practical, one-time bridge for legacy contracts rather than as a permanent feature of the SOFR regime going forward.

10. SOFR Averages and Other Published Conventions

Beyond Term SOFR and compounded SOFR in arrears, the New York Fed also publishes 30-, 90- and 180-day SOFR Averages and a SOFR Index, both compounding the daily SOFR fixings over the stated trailing window, published each business day. These trailing averages are used less often in new-issue credit agreements than Term SOFR or a bespoke compounded-in-arrears calculation with its own lookback and lockout provisions, but they appear in some legacy fallback language and in certain floating-rate note indentures, and a reader reconciling a facility's actual accrued interest against a published SOFR series should first confirm which of these related-but-distinct SOFR-family conventions that facility's own definitions actually specify, rather than assuming any one of them is the market default - none of the SOFR-family conventions is universal across product types.

11. Hedging Floating-Rate Exposure with SOFR Swaps

A borrower with SOFR-based floating-rate debt that wants fixed-rate certainty typically enters into a pay-fixed, receive-floating SOFR swap referencing the same SOFR-family rate its facility uses, converting its floating debt-service profile into an economically fixed one at the swap's fixed rate. The swap curve an analyst would use to price such a hedge, or to value an existing one, is conceptually the same SOFR-swap curve this engagement's own relative-value work uses as the I-spread benchmark leg for the comparable-bond spread analysis elsewhere in this file - a real point of overlap between the benchmark-rate mechanics described in this note and the fixed-income relative-value mechanics used on the notes themselves, even though the notes are not floating-rate instruments and are not directly hedged this way. A reader who sees the swap curve referenced in two different contexts in this file's documents - once as a hedging benchmark for floating-rate debt, once as a relative-value spread benchmark for fixed-rate comparables - should recognize it as the same curve serving two distinct, unrelated analytical purposes, not as two different data sources that happen to coincide.

12. Practical Checklist for Reading a Facility's Rate Mechanics

Before relying on any statement about a floating-rate instrument's benchmark, confirm: (i) whether the facility's own defined terms specify Term SOFR or compounded SOFR in arrears, and for Term SOFR, which tenor; (ii) whether a SOFR floor applies, and at what level; (iii) whether the facility carries a negotiated credit spread adjustment as an original pricing feature (Section 5 above) as distinct from a legacy ISDA/ARRC fallback spread adjustment (Section 4 above) - the two are easy to conflate but apply in entirely different circumstances; (iv) the current Applicable Margin under the facility's own pricing grid, which is a function of the borrower's own leverage or ratings level, not a market-wide constant; and (v) whether the instrument in question is even floating-rate at all - a fixed-rate instrument, like the target's own senior notes, has no SOFR exposure whatsoever, and none of the mechanics in this note changes how that instrument is priced, discounted or hedged.

13. Comparable Transitions in Other Currencies

The USD LIBOR-to-SOFR transition was the largest, but not the only, benchmark reform of its kind. Sterling LIBOR transitioned to SONIA (the Sterling Overnight Index Average, administered by the Bank of England), euro LIBOR and EURIBOR-adjacent reform work centered on Euro Short-Term Rate (EUR STR, administered by the European Central Bank), Swiss franc LIBOR transitioned to SARON, and yen LIBOR transitioned to TONA (the Tokyo Overnight Average Rate). Every one of these replacement rates shares SOFR's basic design philosophy: an overnight, nearly risk-free rate anchored in real, observable transactions rather than a panel-bank submission process, reflecting a common regulatory conclusion, reached independently across jurisdictions, that a submission-based interbank rate had become structurally vulnerable to exactly the kind of manipulation and thin-market unreliability that originally prompted the global LIBOR reform effort. A cross-currency facility or swap negotiated after each respective transition date should reference the appropriate risk-free rate for each currency leg, never a legacy IBOR rate for any currency past that currency's own cessation date.

14. Documentation Language: Hardwired vs. Amendment Approach

The ARRC published two principal documentation approaches for legacy loan agreements transitioning away from LIBOR. The 'amendment approach' left the legacy LIBOR-referencing language in place but added a mechanism requiring the borrower and administrative agent (or a majority of lenders) to agree on a replacement rate at the time LIBOR actually became unavailable, deferring the specific replacement-rate decision to that later date. The 'hardwired approach' instead specified, at origination, a fixed waterfall of successor rates (Term SOFR, then compounded SOFR in arrears, then a further fallback) that would apply automatically and mechanically upon a specified trigger event, with no further lender consent required. Facilities originated after the transition, including this engagement's own credit facility, are drafted directly against Term SOFR as the primary rate from inception and

therefore do not need either fallback mechanism at all - the hardwired-versus-amendment distinction is a legacy-contract question, relevant to instruments that existed before the transition, not to a facility originated after it.

15. Sources for This Note

The historical and mechanical description in this note draws on publicly available material from the Federal Reserve Bank of New York's SOFR and ARRC program pages, the Alternative Reference Rates Committee's own published recommendations and user guides, CME Group's Term SOFR reference rate documentation, ISDA's IBOR Fallbacks Supplement and associated Protocol materials, and the LIBOR Act (part of the Consolidated Appropriations Act, 2023) together with the Federal Reserve's implementing regulation. None of this note's content is specific to, or derived from, this engagement's own confidential transaction materials; it is general market-reference background, consistent with its stated purpose in the introduction above.

16. Frequently Confused Terms

A short list of pairs that are easy to conflate but mean genuinely different things. Term SOFR vs. compounded SOFR in arrears: both are SOFR-based, but the first is forward-looking and set at the start of a period, the second is backward-looking and only fully known near the end of a period (Section 3 above). ISDA/ARRC fallback spread adjustment vs. a facility's own negotiated credit spread adjustment: the first is a fixed, one-time constant published for legacy LIBOR contracts converting to SOFR; the second is an ordinary, deal-specific pricing term negotiated between a borrower and its lender group, unrelated to any LIBOR fallback (Sections 4 and 5 above). SOFR floor vs. Applicable Margin floor: a SOFR floor sets a minimum for the reference rate itself before the margin is added; a margin floor (less common) would instead set a minimum for the spread component - the two protect against different scenarios and a facility need not have either, one, or both. Hardwired approach vs. amendment approach: two different legacy-contract fallback documentation strategies (Section 14 above), neither of which is relevant to a facility originated directly on Term SOFR from inception.